

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.53_k1)

Metric

$$ds^2 = -e^{(2\nu(r,t))} dt \otimes dt + e^{(2\lambda(r,t))} dr \otimes dr + \left(-\frac{Y(r,t)^2}{\rho^2 - 1} \right) d\rho \otimes d\rho + \rho^2 Y(r,t)^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \rho, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} e^{(\nu(r,t))} \right) dt + \left(-\sqrt{\frac{1}{2}} e^{(\lambda(r,t))} \right) dr$$

$$n = \left(\sqrt{\frac{1}{2}} e^{(\nu(r,t))} \right) dt + \left(\sqrt{\frac{1}{2}} e^{(\lambda(r,t))} \right) dr$$

$$m = \left(\frac{\sqrt{\frac{1}{2}} Y(r,t)}{\sqrt{-\rho^2 + 1}} \right) d\rho + \left(i \sqrt{\frac{1}{2}} \rho Y(r,t) \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{\frac{1}{2}} Y(r,t)}{\sqrt{-\rho^2 + 1}} \right) d\rho + \left(-i \sqrt{\frac{1}{2}} \rho Y(r,t) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{1}{2} \sqrt{2} e^{(-\nu(r,t))} \right) dt + \left(-\frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))} \right) dr$$

$$n = \left(-\frac{1}{2} \sqrt{2} e^{(-\nu(r,t))} \right) dt + \left(\frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))} \right) dr$$

$$m = \left(\frac{\sqrt{2} \sqrt{\rho+1} \sqrt{-\rho+1}}{2Y(r,t)} \right) d\rho + \left(\frac{i \sqrt{2}}{2\rho Y(r,t)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2} \sqrt{\rho+1} \sqrt{-\rho+1}}{2Y(r,t)} \right) d\rho + \left(-\frac{i \sqrt{2}}{2\rho Y(r,t)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{1}{2} \sqrt{2} e^{(-\nu(r,t))} \partial_t - \frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))} \partial_r$$

$$\Delta = -\frac{1}{2} \sqrt{2} e^{(-\nu(r,t))} \partial_t + \frac{1}{2} \sqrt{2} e^{(-\lambda(r,t))} \partial_r$$

$$\delta = \frac{\sqrt{2} \sqrt{\rho+1} \sqrt{-\rho+1}}{2Y(r,t)} \partial_\rho + \frac{i \sqrt{2}}{2\rho Y(r,t)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2} \sqrt{\rho+1} \sqrt{-\rho+1}}{2Y(r,t)} \partial_\rho - \frac{i \sqrt{2}}{2\rho Y(r,t)} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2} e^{(-\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t)}{2Y(r,t)} + \frac{\sqrt{2} e^{(-\nu(r,t))} \frac{\partial}{\partial t} Y(r,t)}{2Y(r,t)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}Y(r,t)}{2Y(r,t)} - \frac{\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}Y(r,t)}{2Y(r,t)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{\sqrt{2}\rho}{4\sqrt{\rho+1}\sqrt{-\rho+1}Y(r,t)} - \frac{\sqrt{2}}{4\sqrt{\rho+1}\rho\sqrt{-\rho+1}Y(r,t)}$$

$$\beta = -\frac{\sqrt{2}\rho}{4\sqrt{\rho+1}\sqrt{-\rho+1}Y(r,t)} + \frac{\sqrt{2}}{4\sqrt{\rho+1}\rho\sqrt{-\rho+1}Y(r,t)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{1}{4}\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t) - \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)$$

$$\gamma = \frac{1}{4}\sqrt{2}e^{(-\nu(r,t))}\frac{\partial}{\partial t}\lambda(r,t) - \frac{1}{4}\sqrt{2}e^{(-\lambda(r,t))}\frac{\partial}{\partial r}\nu(r,t)$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \lambda(r,t)}{2Y(r,t)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial t} \lambda(r,t)}{Y(r,t)} + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \lambda(r,t)}{2Y(r,t)} + \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \nu(r,t)}{2Y(r,t)} \\ & + \frac{e^{(-\lambda(r,t)-\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial r} \nu(r,t)}{Y(r,t)} + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \nu(r,t)}{2Y(r,t)} - \frac{e^{(-2\lambda(r,t))} \frac{\partial^2}{(\partial r)^2} Y(r,t)}{2Y(r,t)} - \frac{e^{(-\lambda(r,t)-\nu(r,t))} \frac{\partial^2}{\partial r \partial t} Y(r,t)}{Y(r,t)} - \frac{e^{(-2\nu(r,t))} \frac{\partial^2}{(\partial t)^2} Y(r,t)}{2Y(r,t)}\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = & -\frac{1}{4} e^{(-2\nu(r,t))} \frac{\partial}{\partial t} \lambda(r,t)^2 - \frac{1}{4} e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} \lambda(r,t) \frac{\partial}{\partial r} \nu(r,t) + \frac{1}{4} e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} \nu(r,t)^2 + \frac{1}{4} e^{(-2\nu(r,t))} \frac{\partial}{\partial t} \lambda(r,t) \frac{\partial}{\partial t} \nu(r,t) \\ & - \frac{1}{4} e^{(-2\nu(r,t))} \frac{\partial^2}{(\partial t)^2} \lambda(r,t) + \frac{1}{4} e^{(-2\lambda(r,t))} \frac{\partial^2}{(\partial r)^2} \nu(r,t) - \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t)^2}{4Y(r,t)^2} + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t)^2}{4Y(r,t)^2} + \frac{1}{4Y(r,t)^2}\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned}\Phi_{22} = & \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \lambda(r,t)}{2Y(r,t)} - \frac{e^{(-\lambda(r,t)-\nu(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial t} \lambda(r,t)}{Y(r,t)} + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \lambda(r,t)}{2Y(r,t)} + \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \nu(r,t)}{2Y(r,t)} \\ & - \frac{e^{(-\lambda(r,t)-\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial r} \nu(r,t)}{Y(r,t)} + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \nu(r,t)}{2Y(r,t)} - \frac{e^{(-2\lambda(r,t))} \frac{\partial^2}{(\partial r)^2} Y(r,t)}{2Y(r,t)} + \frac{e^{(-\lambda(r,t)-\nu(r,t))} \frac{\partial^2}{\partial r \partial t} Y(r,t)}{Y(r,t)} - \frac{e^{(-2\nu(r,t))} \frac{\partial^2}{(\partial t)^2} Y(r,t)}{2Y(r,t)}\end{aligned}$$

$$\begin{aligned}\Lambda = & \frac{1}{12} e^{(-2\nu(r,t))} \frac{\partial}{\partial t} \lambda(r,t)^2 + \frac{1}{12} e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} \lambda(r,t) \frac{\partial}{\partial r} \nu(r,t) - \frac{1}{12} e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} \nu(r,t)^2 - \frac{1}{12} e^{(-2\nu(r,t))} \frac{\partial}{\partial t} \lambda(r,t) \frac{\partial}{\partial t} \nu(r,t) + \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \lambda(r,t)}{6Y(r,t)} \\ & + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \lambda(r,t)}{6Y(r,t)} + \frac{1}{12} e^{(-2\nu(r,t))} \frac{\partial^2}{(\partial t)^2} \lambda(r,t) - \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \nu(r,t)}{6Y(r,t)} - \frac{1}{12} e^{(-2\lambda(r,t))} \frac{\partial^2}{(\partial r)^2} \nu(r,t) \\ & - \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \nu(r,t)}{6Y(r,t)} - \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t)^2}{12Y(r,t)^2} - \frac{e^{(-2\lambda(r,t))} \frac{\partial^2}{(\partial r)^2} Y(r,t)}{6Y(r,t)} + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t)^2}{12Y(r,t)^2} + \frac{e^{(-2\nu(r,t))} \frac{\partial^2}{(\partial t)^2} Y(r,t)}{6Y(r,t)} + \frac{1}{12Y(r,t)^2}\end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = & -\frac{1}{6} e^{(-2\nu(r,t))} \frac{\partial}{\partial t} \lambda(r,t)^2 - \frac{1}{6} e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} \lambda(r,t) \frac{\partial}{\partial r} \nu(r,t) + \frac{1}{6} e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} \nu(r,t)^2 + \frac{1}{6} e^{(-2\nu(r,t))} \frac{\partial}{\partial t} \lambda(r,t) \frac{\partial}{\partial t} \nu(r,t) + \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \lambda(r,t)}{6 Y(r,t)} \\ & + \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \lambda(r,t)}{6 Y(r,t)} - \frac{1}{6} e^{(-2\nu(r,t))} \frac{\partial^2}{(\partial t)^2} \lambda(r,t) - \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t) \frac{\partial}{\partial r} \nu(r,t)}{6 Y(r,t)} + \frac{1}{6} e^{(-2\lambda(r,t))} \frac{\partial^2}{(\partial r)^2} \nu(r,t) \\ & - \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t) \frac{\partial}{\partial t} \nu(r,t)}{6 Y(r,t)} + \frac{e^{(-2\lambda(r,t))} \frac{\partial}{\partial r} Y(r,t)^2}{6 Y(r,t)^2} - \frac{e^{(-2\lambda(r,t))} \frac{\partial^2}{(\partial r)^2} Y(r,t)}{6 Y(r,t)} - \frac{e^{(-2\nu(r,t))} \frac{\partial}{\partial t} Y(r,t)^2}{6 Y(r,t)^2} + \frac{e^{(-2\nu(r,t))} \frac{\partial^2}{(\partial t)^2} Y(r,t)}{6 Y(r,t)} - \frac{1}{6 Y(r,t)^2}\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"